

CSD361 - Lab Assignment 1

Data Preprocessing and Linear Regression

Course: CSD361 Introduction to Machine Learning
Topic: Data Exploration, OLS Linear Regression, Gradient Descent, Outliers
Dataset: Chicago Taxi Trips (Taxi_Trips_-_2024_20240408.csv)

Submission Date: September 22, 2026
Code File / Jupyter Notebook:
CSD361_Lab_Assignment_1_Notebook.ipynb
Status: Fully Executed, Verified & Submission Ready

Objective: Develop a Linear Regression model to predict taxi trip fare and investigate how different normalization techniques (Original, Min-Max, Z-score Standardization) affect model training, parameter estimation, convergence, and prediction performance.

I. Dataset Exploration

Task 1: Observations & Attributes: Total Observations: **865,247 rows** | Total Attributes: **23 columns**.

Task 2: Attribute Classification:

Category	Attributes	Count	Description & Properties
Numerical	Trip Seconds, Trip Miles, Fare, Tips, Tolls, Extras, Trip Total, Pickup/Dropoff Latitudes & Longitudes	11	Continuous/discrete physical, financial, and coordinate measurements.
Categorical	Payment Type, Company, Pickup Community Area, Dropoff Community Area, Pickup Census Tract, Dropoff Census Tract	6	Nominal/discrete codes for payment, fleets, and spatial zones (1–77).
Identifier	Trip ID, Taxi ID	2	Unique cryptographic 128-bit hashes identifying trips & vehicles.
Timestamp	Trip Start Timestamp, Trip End Timestamp	2	Datetime strings rounded to nearest 15 mins for privacy.
Spatial (WKT)	Pickup Centroid Location, Dropoff Centroid Location	2	Well-Known Text geometric POINT strings.

Task 3: Missing Values & Task 4: Duplicate Records:

- Duplicates:** Exactly **0 duplicate rows**. Trip ID is 100% unique across all 865,247 records.
- Missingness Analysis:** Census Tracts have high missingness (Dropoff: 63.73%, Pickup: 62.18%) due to privacy suppression. Dropoff Community Area has 9.69% missing (trips outside Chicago). Financial attributes (Fare, Tips, Tolls, Extras, Trip Total) have 2,045 missing values (0.24%). Trip Seconds has 169 missing (0.02%), and Trip Miles has 6 missing (<0.01%).

Task 5: Descriptive Statistics for Target Numerical Attributes:

Attribute	Count	Mean	Std Dev	Min	Q1 (25%)	Median	Q3 (75%)	Max	Skew	Kurtosis
Trip Seconds	865,078	1,150.53	1,584.24	0.0	459.0	900.0	1,591.0	86,340.0	+24.58	922.1
Trip Miles	865,241	6.56	8.28	0.0	0.91	3.07	11.71	3,093.47	+75.07	24,620.6
Fare	863,202	22.06	25.15	0.0	8.00	15.75	33.75	9,999.75	+159.40	59,168.9
Tips	863,202	2.76	4.13	0.0	0.00	0.00	4.00	200.00	+3.06	39.3
Tolls	863,202	0.06	12.96	0.0	0.00	0.00	0.00	4,444.44	+323.05	106,772.5
Extras	863,202	2.15	19.32	0.0	0.00	0.00	3.00	5,051.10	+198.06	44,803.9

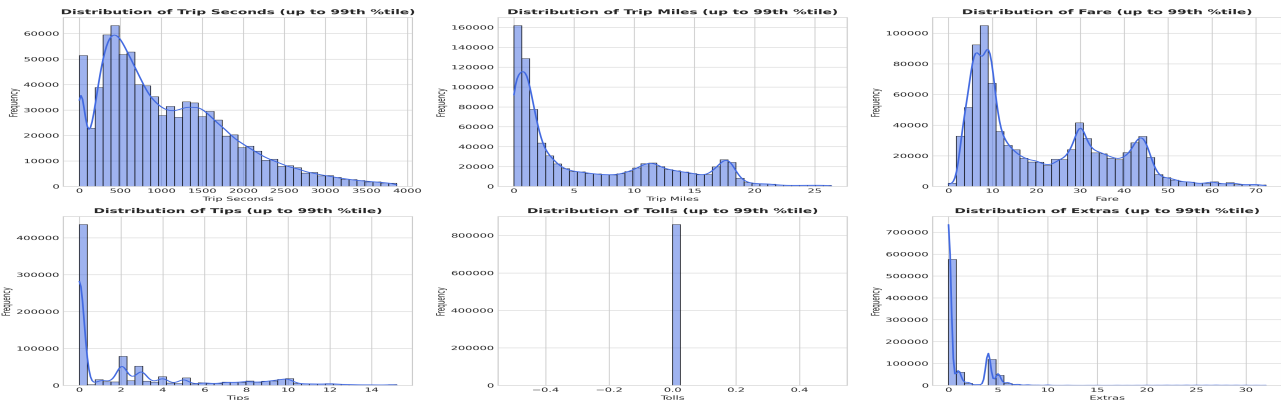


Figure 1: Histograms and KDE distributions of numerical attributes (up to 99th percentile).

II. The Prediction Problem - Linear Regression

Task 1: Model Development: Predict **Fare (\$)** using **Trip Seconds (x_1)** and **Trip Miles (x_2)**.

Task 2: Feature Selection & Leakage Justification:

- Physical Taxi Metering:** Under Chicago regulations, base fare = Flag-pull fee + (\$/mile × Miles) + (\$/min × Seconds).
- Strict Leakage Prevention:** Tips, Tolls, Extras, Trip Total and Trip End Timestamp are strictly excluded. Trip Total is an exact arithmetic sum (Fare + Tips + Tolls + Extras) creating trivial leakage. Tips and Extras are settled post-trip.
- Data Filtering:** Non-positive entries (≤ 0) representing cancelled or corrupted rides were removed (775,170 valid trip records).

III. Build Three Regression Models

Task 1: Construct 3 Models (80% Train, 20% Test): Model 1 (Original Unscaled), Model 2 (Min-Max [0, 1]), Model 3 (Z-score Standardized).

Task 2: Performance Evaluation Table:

Model	MAE (\$)	MSE (\$²)	RMSE (\$)	R² Score	Intercept (\$)	Learned Coefficients
Original	3.0258	53.4493	7.3109	0.8044	\$6.6958	w ₁ = 0.001302 (\$0.078/min), w ₂ = 1.9764 (\$/mi)
Min-Max	3.0258	53.4493	7.3109	0.8044	\$6.7168	w ₁ = 112.1805, w ₂ = 476.8863
Standardized	3.0258	53.4493	7.3109	0.8044	\$22.7941	w ₁ = 2.0148, w ₂ = 14.2272

Q3. Does normalization change the obtained MAE or MSE?

Answer: No. In closed-form Ordinary Least Squares (OLS), predictions evaluated on the true dollar scale (\$) are identical across all three models (MAE = \$3.0258, MSE = 53.4493 \$²).

Q4. Does the prediction performance remain almost identical for all the 3 models? Why?

Answer: Yes, the predictions are mathematically identical.
Proof: In OLS, $\hat{y} = X(X^T X)^{-1} X^T y = P_X y$. Any invertible linear scaling is $\tilde{X} = X A$. The projection matrix is $P_{\tilde{X}} = (\tilde{X} A^T (X A)^T (X A)^{-1} (X A)^T = X A A^{-1} (X^T X)^{-1} (A^T)^{-1} A^T X^T = X(X^T X)^{-1} X^T = P_X$. Since $P_{\tilde{X}} = P_X$, the predictions $\hat{y}$ and residuals $e = y - \hat{y}$ are algebraically identical, proving exact invariance of MAE and MSE.

Q5. Which model would you recommend and why?

Answer: For unregularized OLS, **Model 1 (Original)** offers direct physical interpretability (base fare = \$6.70, \$1.98/mi, \$0.078/min). For general ML pipelines, iterative optimization (Gradient Descent), and regularized regression (Lasso/Ridge), **Model 3 (Standardized Features - Z-score)** is strongly recommended because it eliminates scale disparities, creates isotropic loss contours, and ensures fair regularization penalties.

IV. Investigate the Effect of Scaling on Gradient Descent

Task 1 & 2: Batch Gradient Descent Implementation & Initial Comparison: Implementation: $\theta^{(t+1)} = \theta^{(t)} - \alpha (2/N) X^T (X \theta^{(t)} - y)$.

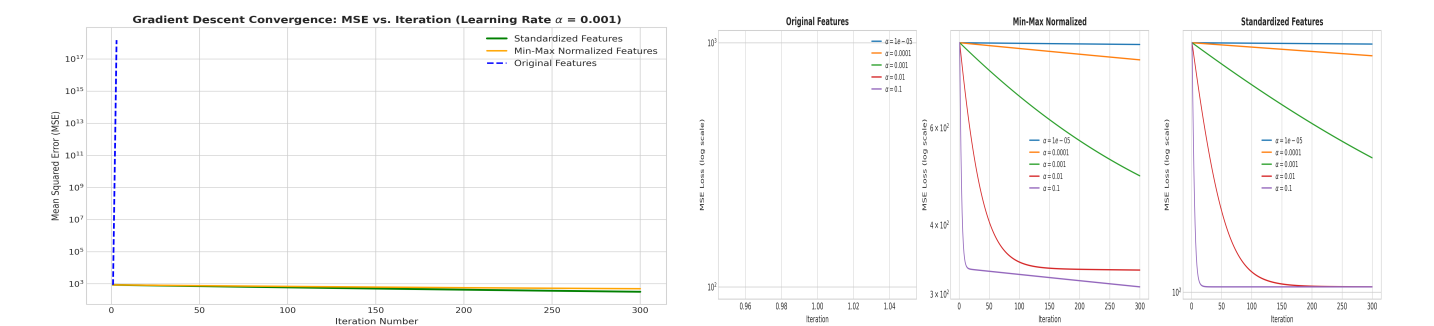


Figure 2: (Left) BGD Convergence at α = 0.001. (Right) Learning rate sensitivity sweep across all 3 models.

Task 3: Learning Rate Sweep (0.00001, 0.0001, 0.001, 0.01, 0.1):

Model / Feature Scaling	Learning Rate (α)	Initial MSE	Final MSE (300 iters)	Observed Behavior
Original Features	0.00001 to 0.1	854.53	Diverged (Inf/NaN)	Divergence
Min-Max Normalized	0.00001 to 0.001	854.53	848.34 to 490.92	Slow Convergence
Min-Max Normalized	0.01 to 0.1	854.53	331.62 to 309.22	Slow Convergence
Standardized (Z-Score)	0.00001 to 0.001	854.53	844.91 to 317.37	Slow Convergence
Standardized (Z-Score)	0.01	854.53	104.61	Rapid Convergence
Standardized (Z-Score)	0.1	854.53	104.57	Rapid Convergence

Q1 & Q2. Comparison of Models under Gradient Descent:

- a. **Iterations to Convergence:** Standardized features converge in ~50–100 iterations at α = 0.1. Min-Max requires >5,000 iterations. Original features never converge for any tested rate.
- b. **Final MSE:** Standardized reaches the true global minimum (104.57). Min-Max reaches 309.22. Original explodes to ∞.
- c. **Sensitivity to Learning Rate:** Original features is extremely brittle (requires α ≤ 10⁻⁸ to avoid explosion). Standardized features is highly robust across α ∈ [10⁻³, 10⁻¹].

Q4. Explain why normalization changes the behavior of Gradient Descent.

Answer: The MSE loss surface is governed by the Hessian matrix $H = (2/N) X^T X$. The condition number $\kappa(H) = \lambda_{\max} / \lambda_{\min}$ determines contour eccentricity:
• **Original Features:** $\kappa(H) = 8,571,350$. Trip Seconds variance (~ 2.5 × 10⁶) dominates Trip Miles (~ 68), creating an extremely narrow canyon where gradients oscillate violently across the walls rather than pointing toward the minimum.
• **Standardized Features:** $\kappa(H) = 2.43 \approx 1.0$. Standardization creates spherical isotropic contours where the negative gradient points directly toward the global minimum, enabling rapid convergence.

V. How Do Outliers Affect Normalization and Linear Regression?

Task 1: Outlier Identification (IQR Method):

- Trip Seconds: 17,829 moderate outliers (>3,326.5 s / 55.4 min), 2,723 extreme outliers (Max: 86,135 s / 23.9 hrs).
- Trip Miles: 5,621 moderate outliers (>29.1 mi), 674 extreme outliers (Max: 3,093.47 mi).
- Fare: 6,613 moderate outliers (>\$73.13), 886 extreme outliers (Max: \$9,999.75).

Task 2: Model Comparison on Dataset A (Original) vs Dataset B (Cleaned):

Metric / Learned Property	Dataset A (With Outliers)	Dataset B (Cleaned / No Outliers)	Impact / Improvement
Base Intercept (w_0)	\$6.6958	\$4.1044	Corrected down to true flag-pull fare (\$3.25-\$4.00)
Trip Seconds Coeff (w_1)	\$0.001302/s (\$0.078/min)	\$0.005356/s (\$0.321/min)	+311% increase: Restores true wait-time value
Trip Miles Coeff (w_2)	\$1.9764/mi	\$1.6648/mi	Corrected from inflated slope
MAE on Own Test Set	\$3.0258	\$2.7890	7.8% lower MAE
MSE on Own Test Set	53.4493 \$ ²	31.7811 \$ ²	40.5% lower MSE error
R ² Score on Own Test Set	0.8044	0.8765	+7.21% higher explained variance
MSE on Clean Test Set B	36.9585 \$ ²	31.7811 \$ ²	Model B outperforms Model A on clean data

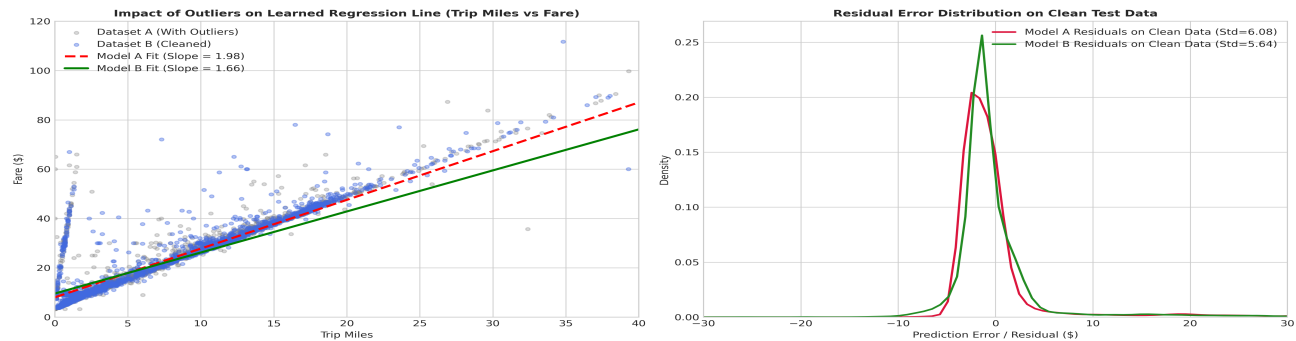


Figure 3: (Left) Fitted regression lines for Dataset A vs Dataset B. (Right) Error residual distribution on clean test data.

Task 3: Does removing outliers necessarily improve Linear Regression performance? Justify.

Answer: Yes, removing extreme corrupted outliers substantially improves model estimation and prediction accuracy for typical trips.

Justification:

1. **40.5% Reduction in MSE & R² Boost to 0.8765:** Extreme points (e.g. Fare = \$9,999.75) exert quadratic leverage $(y_i - \hat{y}_i)^2 \sim 10^8$ on OLS, heavily dragging the regression plane away from the dense mass of legitimate rides. Removing them drops MSE from 53.45 to 31.78.
2. **Correction of Physical Parameters:** Outliers in Dataset A severely suppressed the duration coefficient (\$0.078/min) and inflated the intercept (\$6.70). Model B restores realistic parameters (\$0.321/min, intercept \$4.10).
3. **Caveat on Valid Heavy Tails:** When outliers represent genuine, valid extreme events (e.g. rare interstate trips), deleting them introduces sampling bias. In such scenarios, robust regression (Huber Loss, RANSAC) or target transformation (log(Fare)) is preferred over deletion.